FLINK FORWARD #ASIA 2021 ONLINE 实时即未来 REAL-TIME IS THE FUTURE

流批一体在京东的探索与实践

Exploration and Practice of Unifying Batch and Streaming in JD

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#1 整体思考
Overall Thinking

#2 技术方案及优化
Technical Solutions and Optimizations

#3 落地案例
Cases

#4 未来规划
Future Plan

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#1 整体思考

Overall Thinking

流批一体认知

Cognition of Unifying Batch and Streaming



落地面临的挑战

Challenges

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数据实时性

端到端数据时延

本质性能问题

兼容“批”

开发层面兼容

调度问题

资源问题

流批混部

弹性伸缩

用户观念

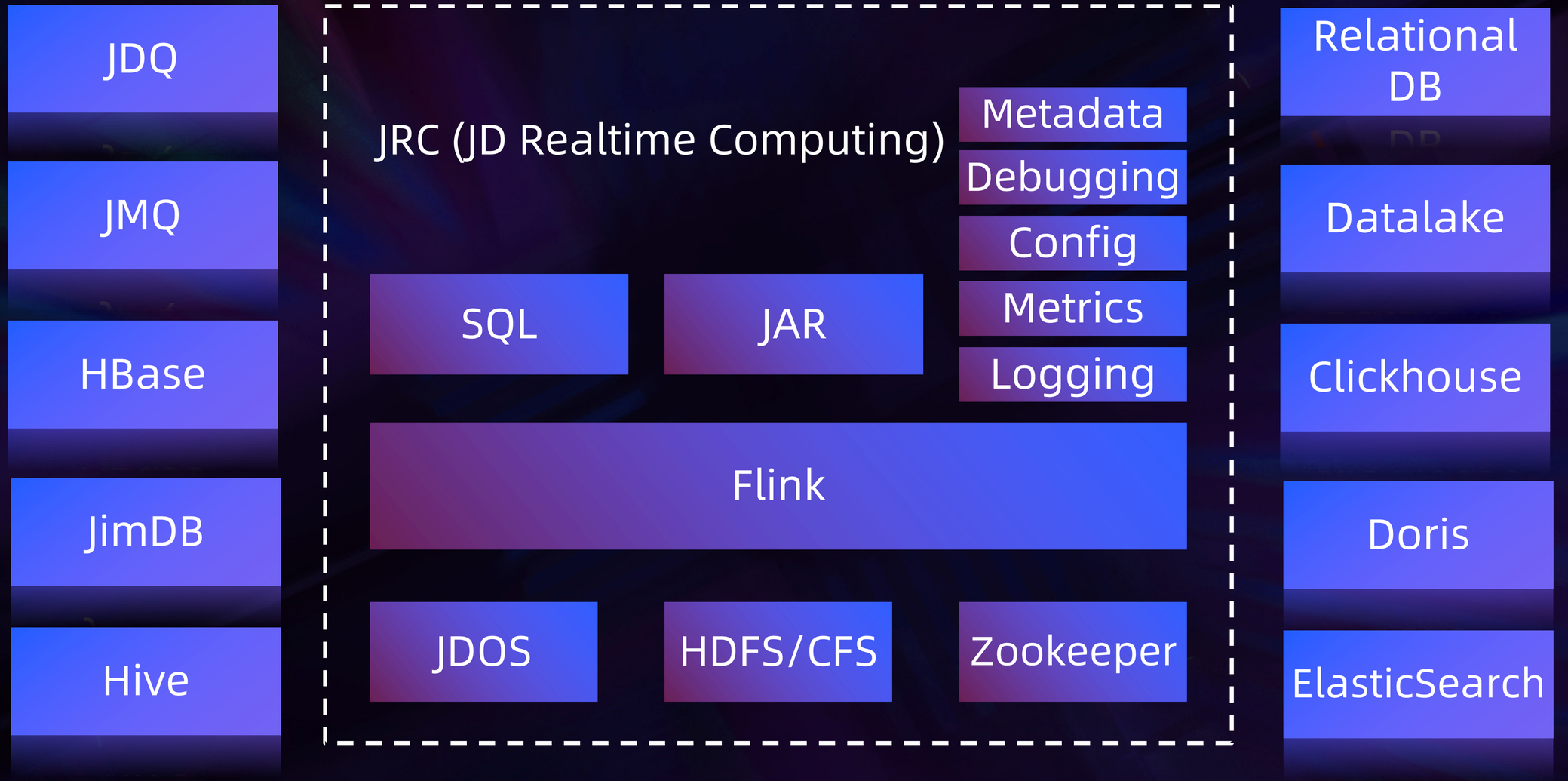
风险与收益

用户视角

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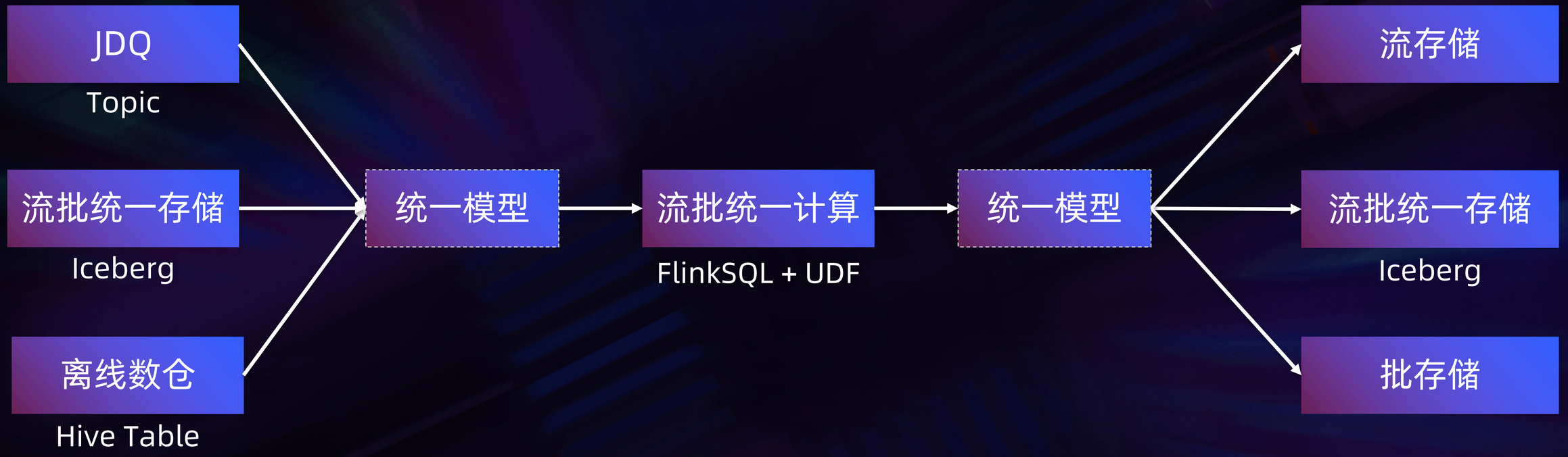
#2 技术方案及优化

Technical Solutions and Optimizations


JDQ
JMQ
HBase
JimDB
Hive
JRC (JD Realtime Computing)
Metadata
Debugging
Config
Metrics
Logging
SQL
JAR
Flink
JDOS
HDFS/CFS
Zookeeper
Relational DB
Datalake
Clickhouse
Doris
ElasticSearch

技术方案-整体架构

Overall Architecture



技术方案-兼容“批”

Compatibility of Batch Processing

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Hive Table

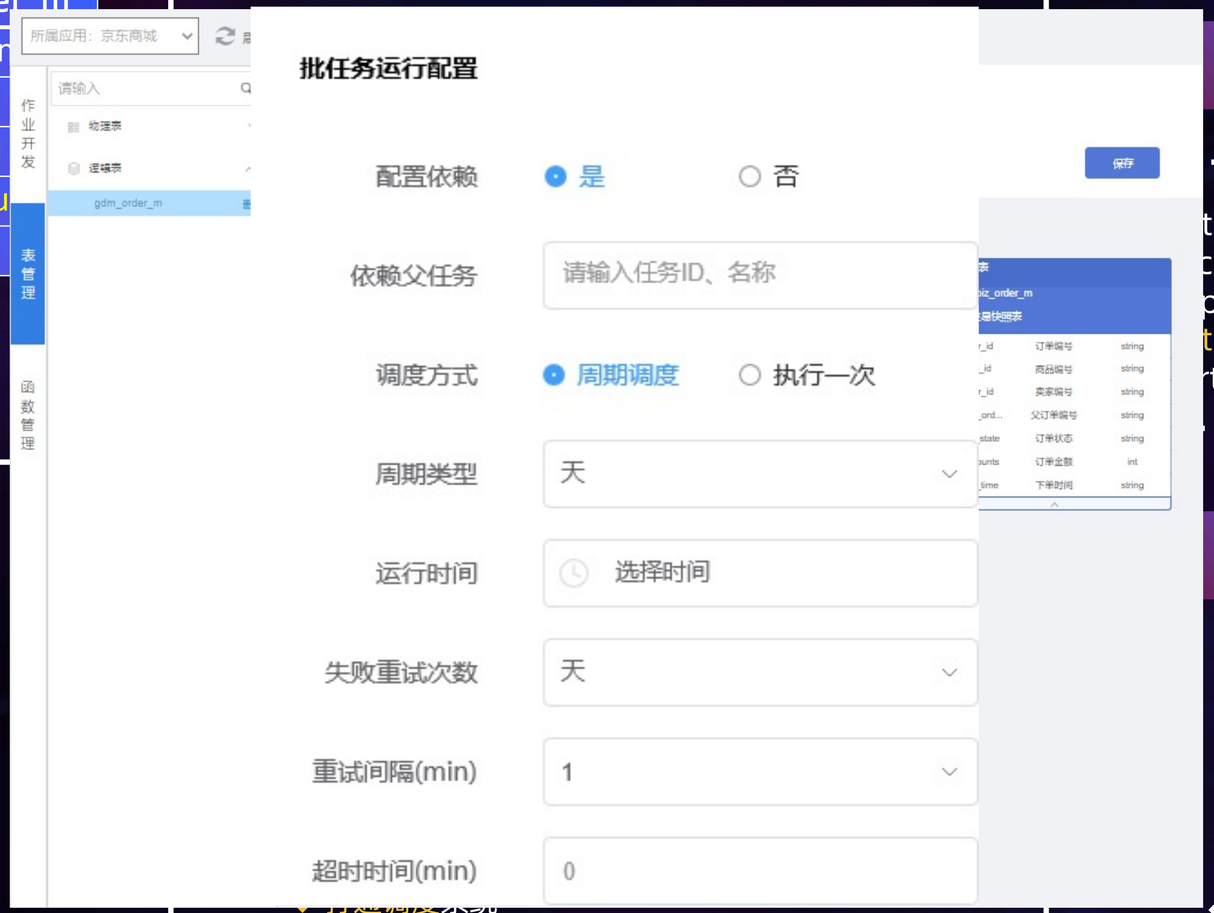
gdm_biz_order_m
order_id
item_id
seller_id
o_amounts
...

Topic

gdm_rt_order_m
order_id_r
item_id
seller_id
order_amou
...

gdm_order_m
order_id
item_id
seller_id
order_amounts
...

FlinkSQL + UDF



upload UDF jar

te catalog xxx
catalog xxx
o function xxx)
te function xxx
rt into xxx（call function）

Hive Metastore

ExtFunctionModule

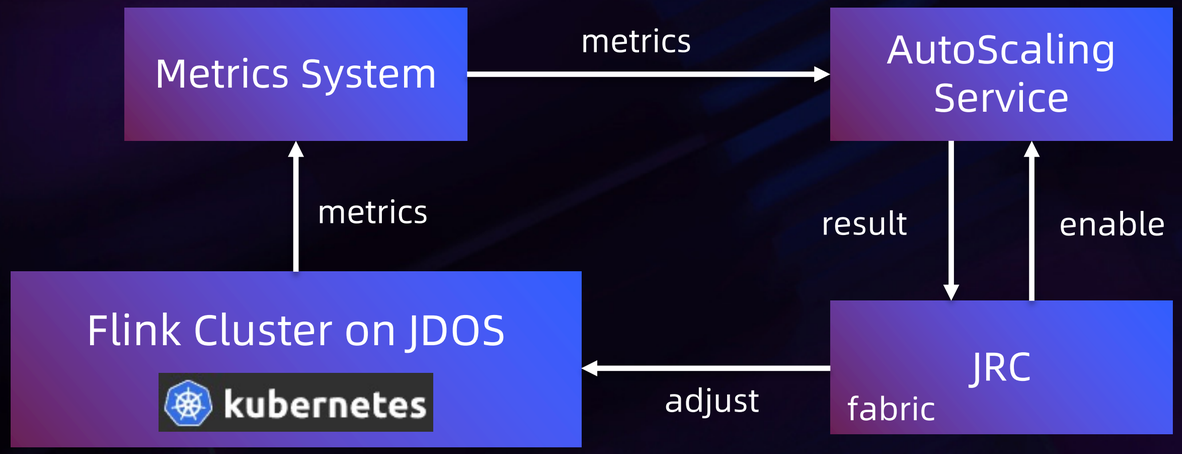
```
ExtFunctionModule
  ExtFunctionModule()
  listFunctions(): Set<String> ↑Module
  getFunctionDefinition(String): Optional<FunctionDefinition> ↑Module
  registerFunction(String, FunctionDefinition): void
  BUILT_IN_FUNC_BLACKLIST: Set<String> = Collections.unmodifiableSet(...)
  tempFunctions: Map<String, FunctionDefinition> = new LinkedHashMap<>()
  instance: ExtFunctionModule = new ExtFunctionModule()
```

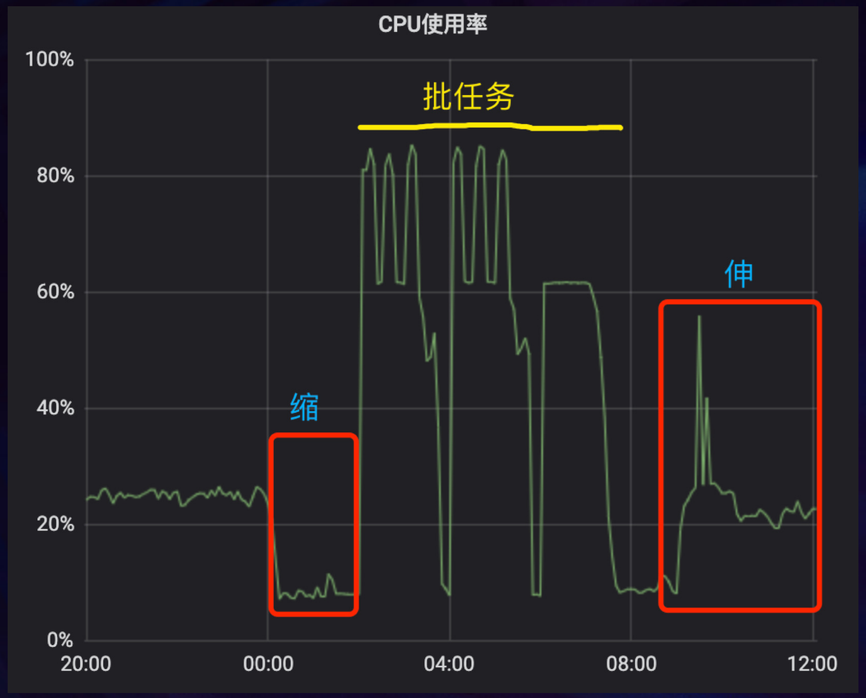
- ◆ 统一模型解决字段映射问题
- ◆ SQL开发面向统一模型层
- ◆ 支持Flink批任务作为数据加工环节
- ◆ 支持自定义Hive UDF、UDAF、UDTF复用
- ◆ 临时注册Function

技术方案-混部及弹性

Hybrid Deploy and Auto Scaling

- 计算资源占用天然错峰(0-8点，流低峰批高峰)
- 流批任务混部的JDOS Zone
- 统一Flink引擎 + 自动弹性能力





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#2 技术方案及优化

Technical Solutions and Optimizations

维表优化

Join优化

Window优化

Iceberg Connector优化

维表优化 - Rebalance

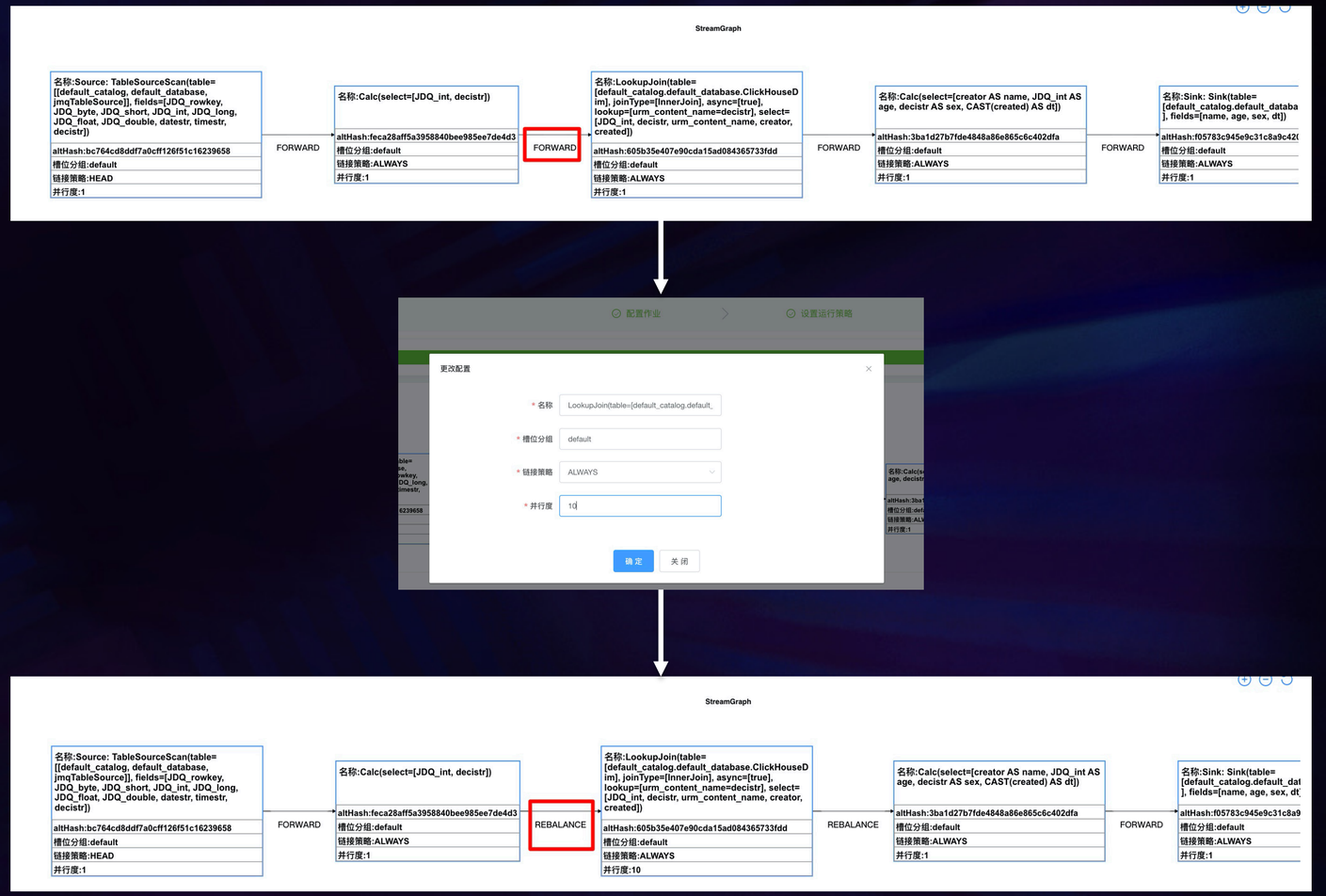
Dimension Table Optimization - Rebalance

依赖平台预览拓扑功能

通过并行度调整实现Forward -> Rebalance

Rebalance -> Dynamic Rebalance

◆ apus.dynamic.rebalance.enable = true



维表优化 - Keyby

Dimension Table Optimization - Keyby

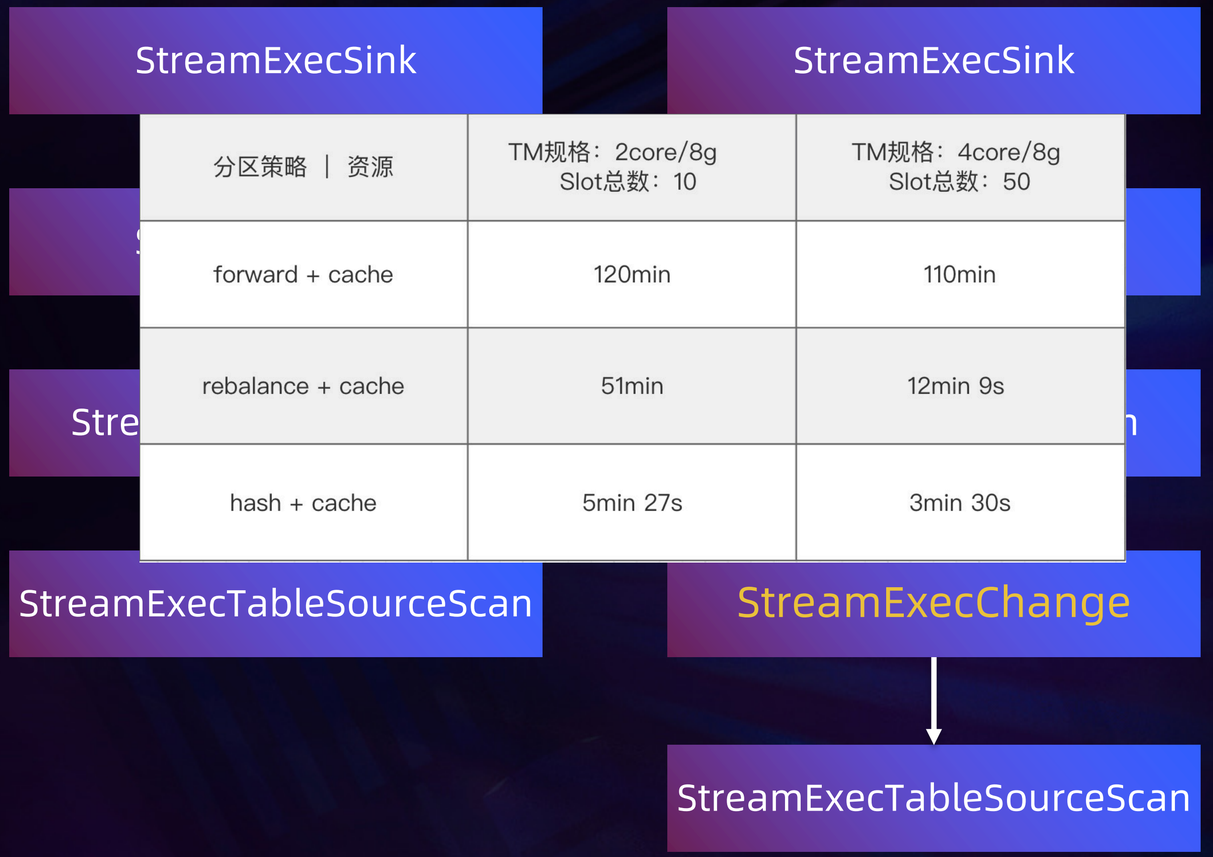
需求场景：

- 如何基于维表缓存优化提高缓存命中率
- 洪峰数据情况下减少数据的缓存淘汰操作

方案：

- Join Key作为Hash条件
- 增加优化规则StreamExecLookupHashJoinRule
- 添加到FlinkStreamRuleSets#PHYSICAL_REWRITE阶段

```
create table hbaseDimTable（
    user_id bigint,
    d row<category_id bigint>
) with (
   ......
   'lookup.hash.enable'='true'
)
```



分区策略 | 资源	TM规格：2core/8g Slot总数：10	TM规格：4core/8g Slot总数：50
forward + cache	120min	110min
rebalance + cache	51min	12min 9s
hash + cache	5min 27s	3min 30s

维表优化 - Mini-Batch

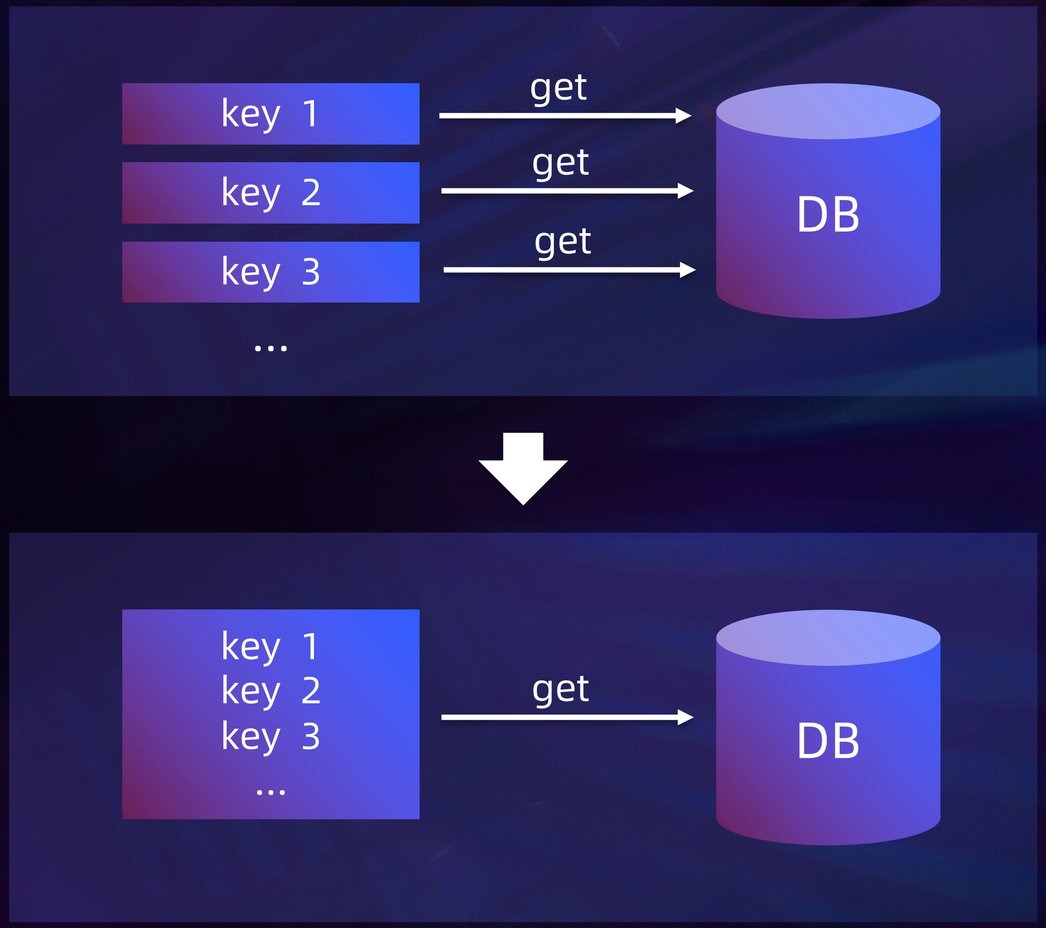
Dimension Table Optimization - Mini-Batch

需求场景：

- 单条记录查询效率低下
- AsyncIO无法解决访问量高的问题

方案：

```
create table hbaseDimTable（
    user_id bigint,
    d row<category_id bigint>
) with (
   ......
   'lookup.async.batch.size'='500',
   'lookup.async.batch.linger'='2s'
)
```



Join优化 - Interval join状态关联优化

Join Optimization - State Optimization of Interval Join

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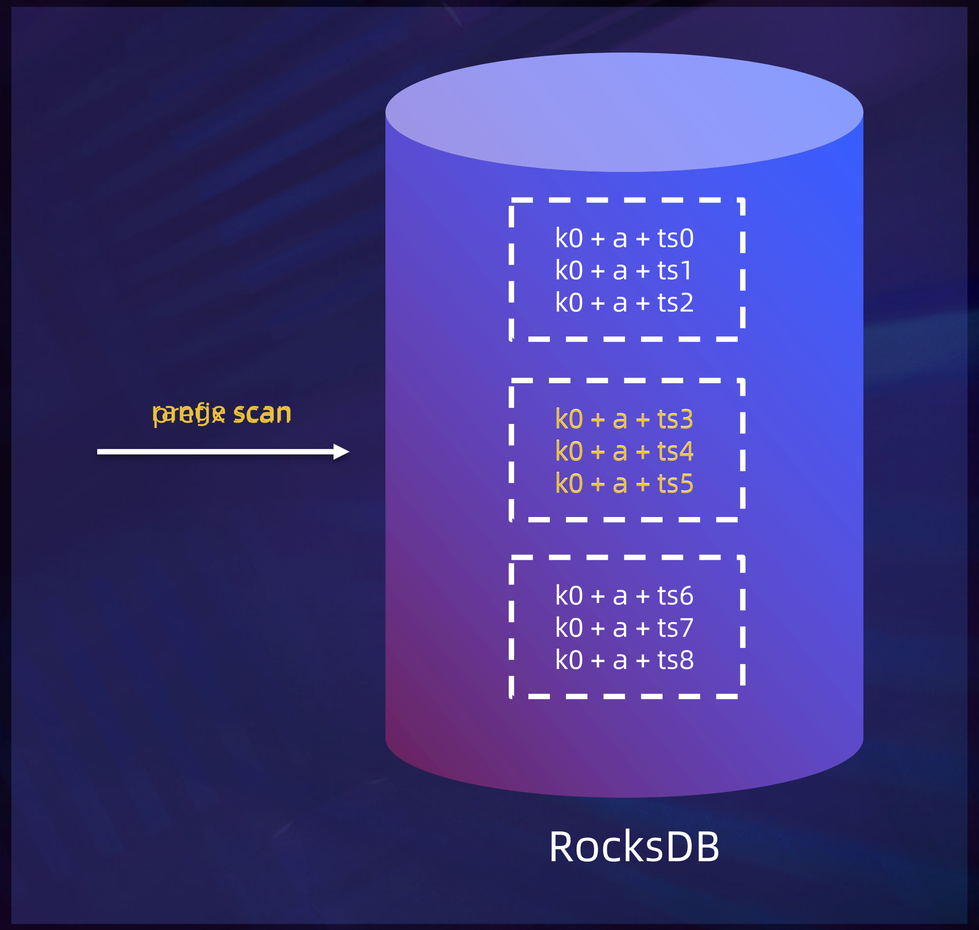
需求场景：

- 高流量Interval join场景大状态
- RocksDB作为Statebackend性能下降
- Flink Job吞吐下降，发生背压

方案：

精确构建数据扫描边界

- keyGroup + joinKey + namespace + timestamp[lower, upper]



Join优化 - Regular inner join大状态

Join Optimization - Huge State Optimization of Regular Join

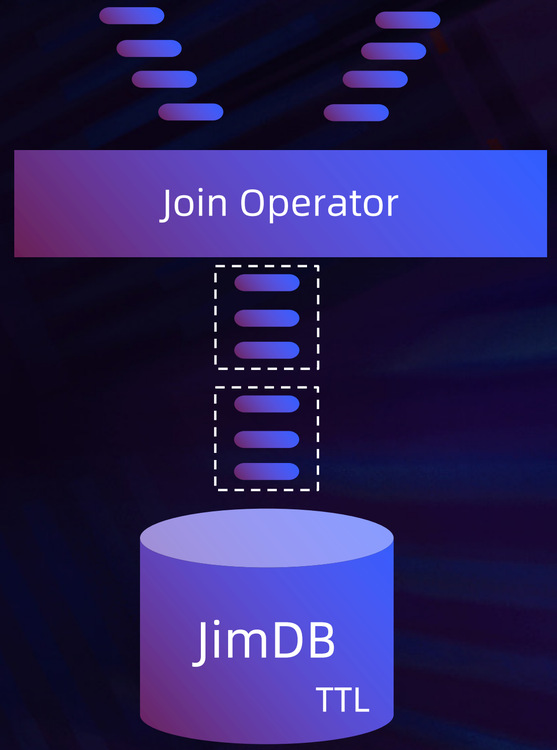
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需求场景：

- 实时流 regular inner join超大状态
- table.exec.state.ttl设置大窗口时间

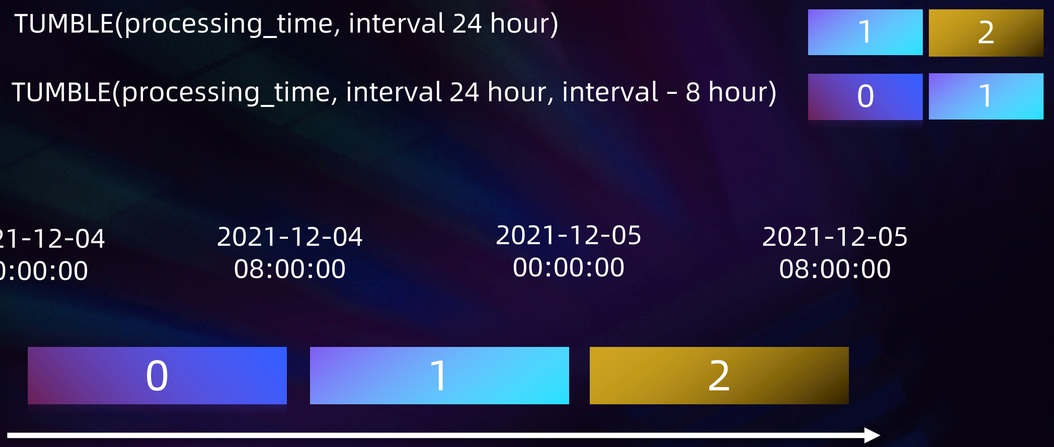
方案：

- Join算子以mini-batch方式将数据存至JimDB
- 借助Jimdb TTL机制实现过期数据清理
- table.join.state.jimdb.enable = true



Window优化 -窗口偏移量 & 增量窗口

Window Optimization - Window Offset and Incremental Window

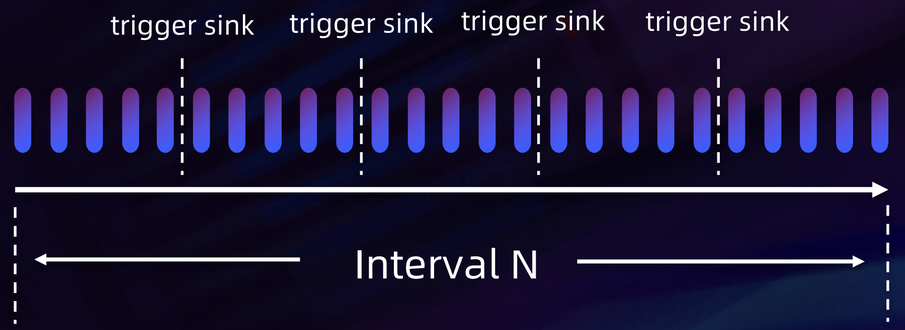


◆解决本地非0时区跨天级别窗口统计错误问题 (1.12 processing time)

◆灵活设置窗口开始结束时间

◆HOP、TUMBLE

◆TUMBLE(time_attr, window_size, window_offsets)



◆根据设置的增量间隔触发，输出窗口的当前计算结果

◆HOP、TUMBLE

◆TUMBLE(time_attr, window_size, window_offsets, trigger_interval)

Iceberg Connector优化 - 小文件合并

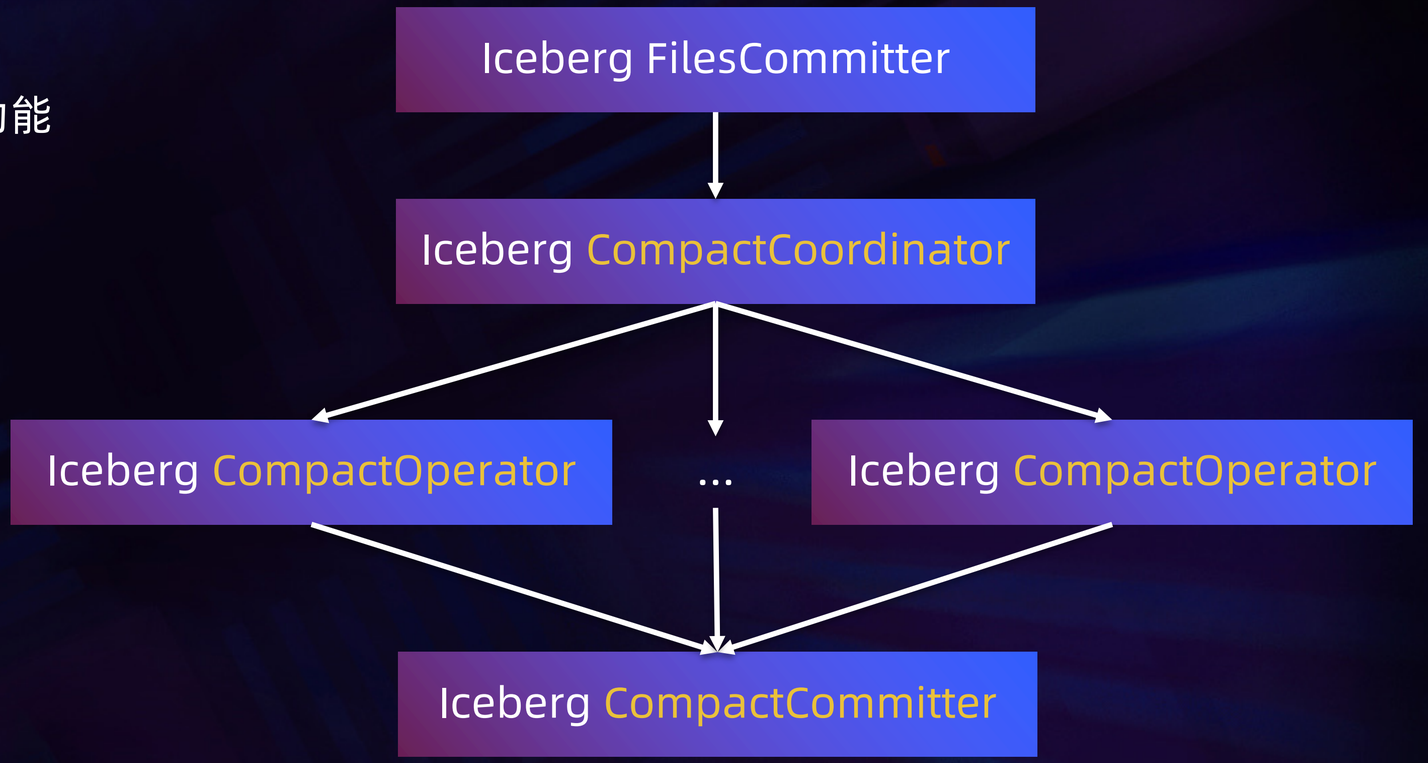
Iceberg Connector Optimization - Merging small files

需求场景：

- Iceberg社区未提供基于Flink算子的小文件合并功能
- 大量小文件影响NameNode性能

方案：

```
create table icebergTable（
    user_id bigint,
    category_id bigint
) with (
   ......
   'auto-compact'='true',
   'compact.delta.commits'='5'
)
```



Iceberg Connector优化 - PruningTableSchema

Iceberg Connector Optimization - PruningTableSchema

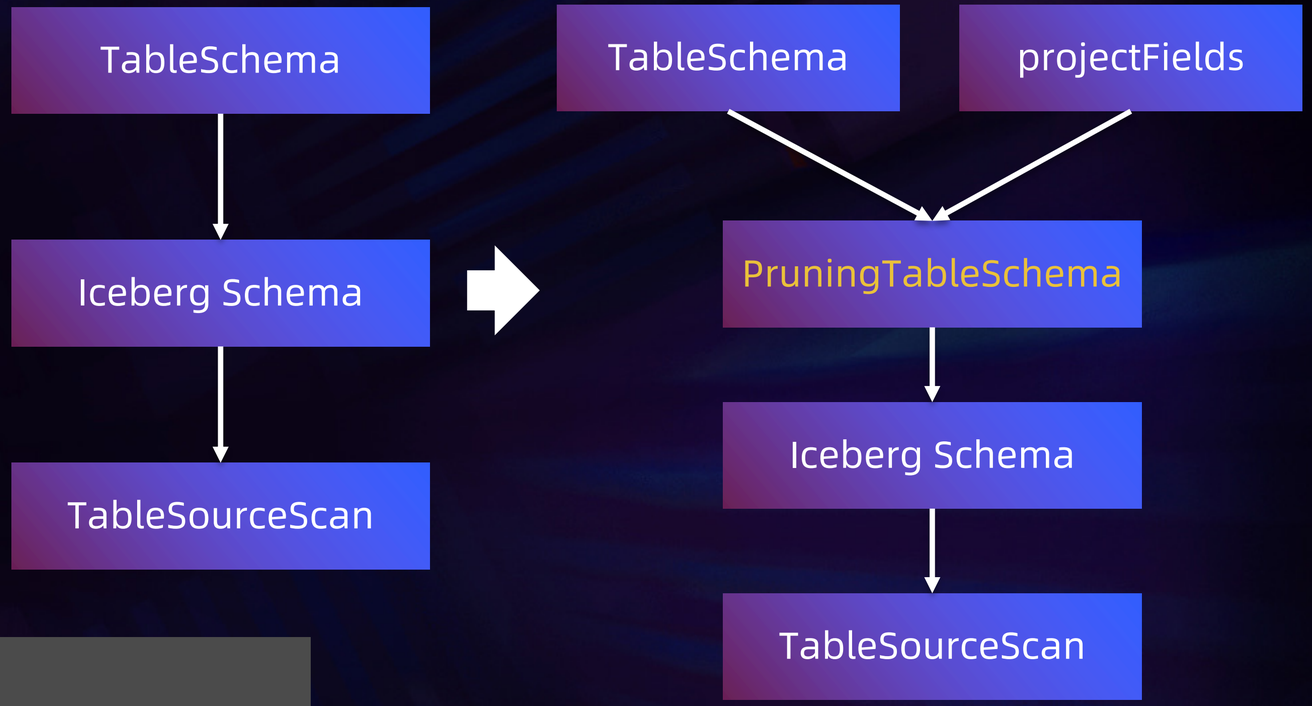
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需求场景：

- Iceberg Connector读取复杂嵌套结构扫描大量无效数据
- CPU、网络带宽负载高

方案：

- 根据查询对嵌套结构进行列裁剪




```
select  brd,
        chancatecd.chanFirstCateCd as chanFirstCateCd,
        base.words.word as word,
        details.infos.j.ju.jum as jum
```

```
root
 |-- brd: STRING
 |-- chancatecd: ROW<`chanFirstCateCd` STRING, `chanSecondCateCd` STRING, `chanThirdCateCd` STRING, `chanFourthCateCd` STRING>
 |-- base: ROW<`words` ROW<`word` STRING, `type` STRING>, `has` ROW<`haslb` STRING, `hasll` STRING, `haslo` STRING, `haslr` STRING>>
 |-- details: ROW<`infos` ROW<`j` ROW<`ju` ROW<`jum` STRING, `juc` STRING, `jus` STRING>>>>
```

```
root
 |-- brd: STRING
 |-- chancatecd: ROW<`chanFirstCateCd` STRING>
 |-- base: ROW<`words` ROW<`word` STRING>>
 |-- details: ROW<`infos` ROW<`j` ROW<`ju` ROW<`jum` STRING>>>>
```

其他优化

Other Optimizations

Interval outer join优化

UDF复用

Keyby语法扩展

维表预加载

Iceberg Connector从指定Snapshot读取

……

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#3 落地案例

Cases

FlinkSQL现状

Current Status of FlinkSQL in Production

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700+

任务数

15%

任务占比

1.1亿+

FlinkSQL任务处理峰值

1.12

Flink版本

实时通用数据层RDDM建设

Real-time Common Data Layer RDDM



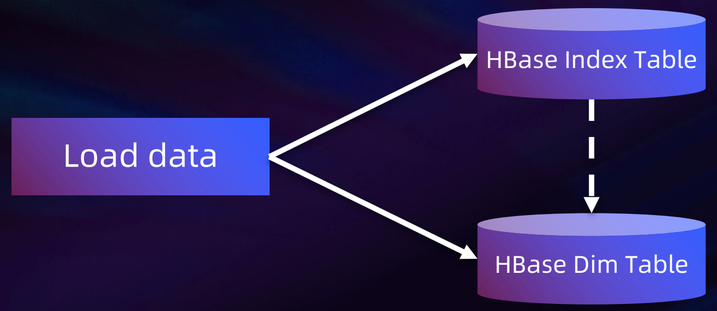
- 服务黄金眼/商智、JDV、广告算法、搜推算法等核心业务
- RDDM已有实时业务模型与离线GDM、ADM层业务加工逻辑一致
- 流量大、实时性要求高、定制化逻辑

实时通用数据层RDDM建设

Real-time Common Data Layer RDDM

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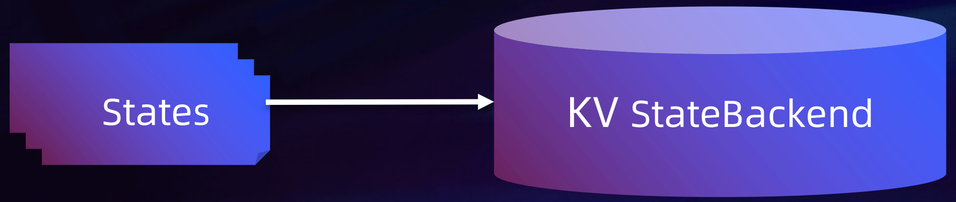
◆ 当前消息需要参与计算的所有相关数据一次性加载进内存



◆ 维表预加载 + 维表Keyby

◆ 定制Connector实现二级索引查询

◆ 部分计算逻辑关联180天的历史数据



◆ Join场景定制优化

◆ 支持KV类型的状态后端

流量买卖黑产舆情分析

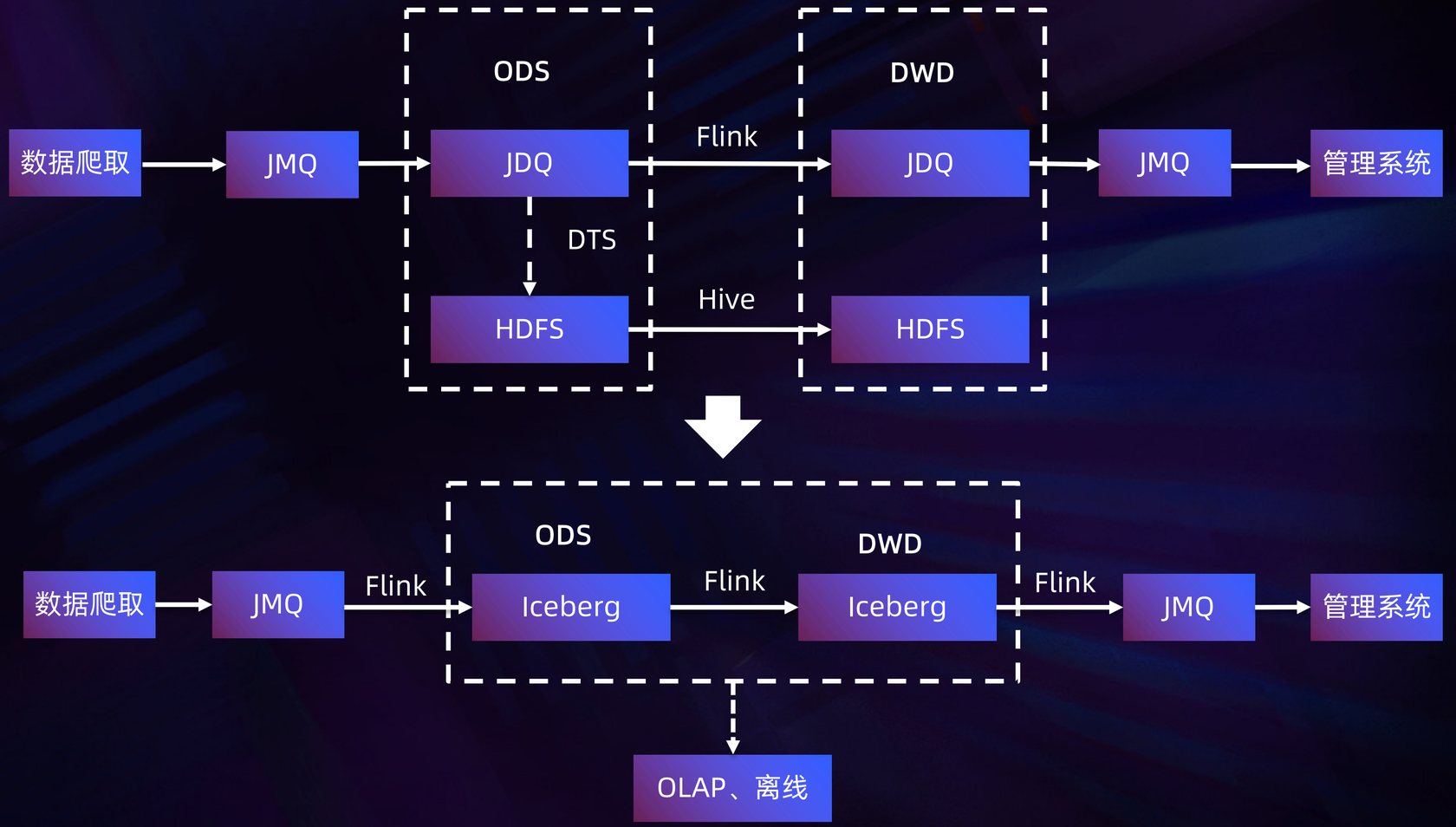
Public Option Analysis of Traffic Deal in Black Market

业务特点：

- 端到端接受分钟级时延
- 离线、实时模型相同

达成效果：

- 端到端时延1分钟左右
- 小文件合并提升性能
- 存储、计算成本显著降低
- 开发、维护成本降低30%+



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#4 未来规划

Future Plan

未来规划

Future Plan

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#1 业务拓展

- FlinkSQL任务推广
- 探索更多流批一体业务场景
- 流批一体产品形态打磨
- 便捷的元数据管理

#2 平台能力

- Join场景优化
- 大状态场景优化
- KV类型的State Backend
- 端到端实时性提升探索


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THANKS